

Factoring

Factoring writes an expression as a product.

$$x^2 + 7x + 12$$

What two binomials could have this product?

Greatest common factor

$$18x^3 + 12x^2$$

What is the greatest common monomial factor?

Factoring out the GCF

$$15x^4 - 10x^3 + 5x^2$$

What completely factored form results?

A common binomial factor

$$4x(x + 3) - 5(x + 3)$$

What factored form uses the repeated binomial?

Factoring by grouping

$$x^3 + 2x^2 + 3x + 6$$

How can we factor this polynomial by grouping?

Factoring a trinomial

$$x^2 + 9x + 20$$

What is the completely factored form?

A trinomial with a negative constant

$$x^2 - x - 12$$

What is the completely factored form?

A trinomial with two negative factors

$$x^2 - 11x + 24$$

What is the completely factored form?

A non-monic trinomial

$$6x^2 + 11x + 3$$

What is the completely factored form?

A non-monic trinomial

$$8x^2 - 2x - 3$$

What is the completely factored form?

Difference of squares

A difference of squares factors into conjugate binomials.

$$a^2 - b^2 = (a - b)(a + b)$$

How would we factor 25x squared minus 9?

A repeated factoring step

$$x^4 - 81$$

What is the completely factored form over the integers?

Perfect square trinomials

$$x^2 - 14x + 49$$

What binomial square equals this expression?

A perfect square with coefficients

$$9x^2 + 24x + 16$$

What binomial square equals this expression?

A sum of cubes

$$x^3 + 27$$

What is the factored form over the integers?

A difference of cubes

$$8x^3 - 125$$

What is the factored form over the integers?

A common factor before a pattern

$$3x^3 - 48x$$

What is the completely factored form over the integers?

Factoring with fractional exponents

Assume x is nonnegative.

$$x^{3/2} + 4x^{1/2}$$

What expression results when we factor out the smallest power of x ?

Factoring with negative exponents

Assume x is nonzero.

$$x^{-3} + 5x^{-2}$$

What expression results when we factor out the smallest power of x ?

Checking a factorization

$$x^2 - 2x - 15 = (x - 5)(x + 3)$$

How can we verify this proposed factorization?

The coefficient domain

$$x^2 + 9$$

Can we factor this polynomial into real linear factors?

A factoring strategy

Factor over the integers.

$$2x^4 - 32$$

Which method should we use first, and what is the completely factored form?